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Narappa G
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Raghavendra Yaligar
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Jyothi R
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Radha J
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Kavitha Ullikashi
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Shruthi N
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Corresponding Author:
Narappa G
ICAR-Krishi Vigyan Kendra, Gangavthi (Koppal), University of Agricultural Sciences, Raichur, Karnataka, India

Drone application: Perspectives, benefits and challenges under Tungabhadra command area of Koppal district

Narappa G, Raghavendra Yaligar, Jyothi R, Radha J, Kavitha Ullikashi and Shruthi N

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Abstract
The adoption of drone technology in agriculture has gained traction in recent years, offering precision, efficiency, and cost-effectiveness. the scientists of ICAR-Krishi Vigyan Kendra, Gangavathi (Koppal district) Under University of Agricultural Sciences, Raichur Karnataka state were conducted a survey on drone application usage during 2024-25. This survey examines the use of drones for chemical spraying in different crops within the Tungabhadra Command Area (TCA). It explores the perceived Perspectives, benefits and challenges of precision technology as experienced by local 50 farmers. In the present survey data was collected through personal interview and it provide insights into its effectiveness, economic viability, and environmental impact, offering a comprehensive evaluation of its role in modern agriculture. The survey reported that 92% of the farmers were expressed that by adapting drone technology has increased the yield by 5 to 10 per cent in paddy, Sorghum and pigeonpea. Further the survey reveals that drone technology usage has enhancing their income by five per cent.

Keywords: Drones, paddy cultivation, chemical spraying, Tungabhadra command area, farmers' opinions, knowledge, benefits, and limitations

Introduction

The continuous growth in agriculture in terms of increased productivity and production is very much essential to meet the needs of ever-growing population in India. The Green Revolution has been the cornerstone of India's agricultural achievement transforming the country from food deficient to self-sufficient during the late 1970s. Thereafter, the agriculture sector in India has been successful in keeping pace with ever increasing food demand of growing population. Agriculture, a crucial sector for global food security, is increasingly relying on technological innovations to enhance productivity and sustainability (Murugan *et al.*, 2024) [2]. Although the country has achieved self-sufficiency in food production, the living conditions of the farmers is declining year after year. Further, it has been found that the large chunk of farmers want to leave farming and look for opportunities in non-agriculture sectors due to the reasons like climate change, explosive population, land fragmentation, shortage of resources and shortage of agricultural labour force *etc* (Singh and Prabhsimran, 2023) [3]. In these circumstances, science holds the solution leading to the emergence of advanced methods like drone technology. While drones have long been utilized by farmers, their integration into agriculture is still at an early stage. By leveraging drone capabilities India's agriculture sector has the potential to overcome limitations and achieve levels of innovation and prosperity.

Combination of drone technology in agriculture has emerged as an innovative development, offering innovative solutions to the challenges faced by the farming community. Traditional farming methods, while effective, often lack the precision and real-time data necessary to optimize yields and ensure sustainability. Drones, equipped with advanced sensors and imaging technologies, are revolutionizing the way farmers monitor and manage their crops. By providing high-resolution aerial imagery, real-time data analytics, and precise mapping capabilities, drones enable farmers to make more informed decisions, improve crop health monitoring, and optimize the use of resources such as water, fertilizers, and pesticides.

The Tungabhadra Command Area, a major agricultural region in Karnataka, India, predominantly cultivates paddy under irrigated conditions. Traditional methods of pesticide and fertilizer application often involve significant labour, time and environmental risks (Singh and Prabhsimran, 2023) [3]. Drone technology has emerged as an innovative solution for efficient agrochemical application (Murugan *et al.*, 2024) [2]. This paper investigates the role of drones in paddy cultivation, focusing on their advantages and drawbacks based on farmers' experiences in the region. This study seeks to fill the gap by deliberate the current knowledge level of drone technology among the farmers. The objectives of the study is to understand the Perspectives, benefits and challenges among the farmers in Tungabhadra Command Area.

Materials and Methods

The present study was carried out in Koppal district of Karnataka state. Since this district is one of predominate district under Tungabhadra Command Area. Fifty drone technology used farmers were personally interviewed by using a interview schedule employing personal interview method. Ex-post facto research design was adopted for the study and collected data was scored and analyzed using mean and percentage, further limitations of using drone application rank was given by using mean scores of all the statements.

Results and Discussion

The survey results presented in the Table1, indicate farmers' outlook and awareness of drone technology in relation to its uses in agriculture. The majority (80.00%) affirmed of farmers Drones can be used to spray crops precisely and efficiently, which showing a strong responsiveness of drones' role in enhancing precision and efficiency in crop spraying. 14.00 per cent of farmers agreed with statement that A smart farming technique that uses drones to apply fertilizers, pesticides, and other treatments to crops suggesting that a small proportion of respondents are familiar with drones being used for such purposes in farming. Only 3 farmers (6.00%) agreed with drone help to reduce the environmental impact of chemical use which indicating a lower level of understanding of drones' potential environmental benefits in farming. Drones have the potential to reduce the overuse of chemicals by allowing for targeted application, minimizing waste and runoff, and decreasing the overall volume of chemicals usage.

With respect to use of Drone in agriculture, 35 respondents (70.00%) agreed that drones improve the effectiveness of spraying operations, followed by 18 per cent time saving and 12 per cent less labour consumption, suggesting that a large proportion of respondents recognize the value of drones in enhancing the precision and effectiveness of pesticide and fertilizer application (Chiranjeeb *et al.*, 2022) [1].

Table 1: Farmer's Perspectives and Benefits under Tungabhadra Command Area n=50

Sl. No.		Statements		
1		**What is Drone technology?**	**No.**	**Percentage**
	a.	a smart farming technique that uses to apply fertilizers, pesticides, and other treatments to crops	7	14.00
	b.	Drones can be used to spray crops precisely and efficiently,	40	80.00
	c.	It can help to reduce the environmental impact of chemical use	3	6.00
		Total	50	100.00
2		**Use of Drone in agriculture improves**		
	a.	Less labour consumption	6	12.00
	b.	Time saving	9	18.00
	c.	Effective spray	35	70.00
		Total	50	100.00
3		**Drone spray is suitable to**		
	a.	Paddy	25	50.00
	b.	Maize	8	16.00
	c.	Red garm	13	26.00
	d.	Sorghum	4	8.00
		Total	50	100.00
4		**Drone spray can be used in**		
	a.	Agricultural crops	45	90.00
	b.	Horticultural crops	5	10.00
	c.	Forestry	0	00.00
		Total	50	100.00
5		**The Drone spray can be more suitable to**		
	a.	Large farmers	30	60.00
	b.	Medium farmers	16	32.00
	c.	Small farmers	4	8.00
		Total	50	100.00
6		**Drone spray can be more impacted on**		
	a.	Pesticide application	32	64.00
	b.	Fertilizer application	10	20.00
	c.	Herbicide application	5	10.00
	d.	Granules application	3	6.00
		Total	50	100.00
7		**Expected benefits of Drone spray**		
	a.	Yield enhancement	6	12.00
	b.	Labour intensive	13	26.00
	c.	Less hazardous	29	58.00
	d.	Less water usage	2	4.00
		Total	50	100.00
8		**Risk during drone spray**		
	a.	More risk	27	54.00
	b.	Moderate risk	18	36.00

	c.	Less risk	5	10.00
		Total	50	100.00
9		**Drone spray can enhance the yield**		
	A	15 to 20%	0	00.00
	B	10 to 15%	4	8.00
	C	5 to 10%	46	92.00
		Total	50	100.00
10		**Economic effectiveness of the Drone spray**		
	A	More	34	68.00
	B	Medium	13	26.00
	C	Less	2	4.00
		Total	50	100.00

In case of drone spray suitable to different crop, Majority of farmers (50.00%) in the study area consider drone spraying suitable for paddy cultivation, since Tungabhadra Command Area major crop is paddy, so it is the most recognized crop for drone spraying followed by Redgram (26%), Maize (16%) and sorghum (8%). It is clearly indicated that paddy is widely seen as a crop that benefits from drone technology. Paddy fields often have large, uniform areas that make them well-suited for drone spraying, which can apply fertilizers and pesticides more efficiently and uniformly compared to manual methods. This high level of recognition suggests that farmers are either familiar with or are more confident in using drones for paddy cultivation. Here majority (90.00%) of farmers believe that drone spraying can be used for agricultural crops, followed by horticultural crops. Drones are particularly useful for tasks such as spraying pesticides, fertilizers, and herbicides over vast areas with high efficiency and precision.

More than half (60.00%) of the farmers have faith on drone spraying is more suitable for large land holding farmers, it indicating a strong preference for this group, followed by lower recognition for medium farmers (32%) and small farmers (8%) indicates varying perceptions of drone technology's applicability based on farm size and the resources required for drone adoption.

In case of drone impact on different application part, 64 per cent people believe that drone spraying has the most significant impact on pesticide application, followed by fertilizer application (20.00%), herbicide application (10.00%) and granules application (6.00%). Drones can be particularly beneficial in pesticide application due to their ability to spray large areas quickly and efficiently while maintaining precision. They also help in targeting specific areas affected by pests, reducing the overall amount of pesticide used and minimizing environmental impact (Chiranjeeb *et al*., 2022) [1].

The survey results provide insights into the expected benefits of drone spraying as perceived by the respondents. From the survey 29 respondents (58.00%) believe that drone spraying is less hazardous compared to traditional spraying methods, it reflecting a significant recognition of safety benefits of drones spaty can provide. 26.00 per cent of respondent expresses drone spraying with being labour-intensive, followed by yield enhancement (12.00%) believe that drone spraying will lead to enhanced crop yield, only 4.00 per cent respondent think that drone spraying results in less water usage, indicating a relatively low consciousness or belief in drones' role in water conservation. This benefit is particularly important for pesticide and herbicide applications, where traditional spraying methods expose farmers to direct contact with chemicals. The recognition of this benefit reflects growing awareness of the safety advantages of drone technology in reducing health risks associated with manual pesticide application.

The highest percentage (54%) of farmers indicating that drone spraying involves more risk concerns about technical failures, operational accidents, or the safety of chemicals applied via drones. With 36% of respondents perceiving moderate risk, it is clear that many recognize that while drone spraying may carry some inherent risks, these can be mitigated with proper training, equipment maintenance, and precautions. smaller proportion (10%) of respondents perceiving less risk may be more familiar with the technology or have had positive experiences with drone spraying.

In case of yields enhancement, large majority (92.00%) respondents believe that drone spraying can enhance the yield by 5 to 10%, reflecting the most widely held expectation of a moderate yield increase. Only 8.00 per cent respondent expect a yield enhancement of 10 to 15% from drone spraying, indicating a small minority who see moderate yield improvements.

The results suggest that drone spraying is generally viewed as economically effective, with 68% of respondents considering it to offer more economic benefits, 26% seeing it as moderately effective, and 4% considering it less effective. This positive outlook on economic effectiveness reflects the growing recognition of drones' potential to reduce labour costs, improve input efficiency, and increase productivity (Zhang *et al.*, 2022) [6].

Table 2: Limitations in Using Drone application n=50

Sl. No	Statements	More severe		Severe		Not severe		Rank
		F	%	F	%	F	%	
1	Battery related issues	36	72.00	10	20.00	4	8.00	II
2	Lack of literacy among farmers	17	34.00	30	60.00	3	6.00	IV
3	More time and practice required to learn the drone operation	9	18.00	33	66.00	8	16.00	VI
4	Registration procedures	10	20.00	31	62.00	9	18.00	VI
5	Lack of trained man power	5	10.00	10	20.00	35	70.00	XI
6	Clarification is difficult if any doubt arises	5	10.00	15	30.00	30	60.00	IX
7	Initial cost is more	35	70.00	11	22.00	4	8.00	III
8	Recurring expenditure is more	6	12.00	12	24.00	32	64.00	X
9	Lack of training centers	7	14.00	27	54.00	16	32.00	VIII
10	Lack of locally relevant information	2	4.00	12	24.00	36	72.00	XII
11	Volume of spray	15	30.00	25	50.00	10	20.00	V
12	Weather related issues	38	76.00	10	20.00	2	4.00	I
13	Government support and policies	10	20.00	32	60.00	8	16.00	VI

The study explores the limitations faced by the farmers when using drone technology, focusing on the severity of challenges encountered. The data was collected across three categories: "More Severe," "Severe," and "Not Severe." Based on this, the limitations are ranked to highlight the most persistent issues. From the Table 2, Weather-related issues emerged as first rank as the most severe limitation, with 76% of farmers reporting it as "More Severe.", followed by Battery-related issues were identified as the second most severe limitation, with 72% of respondents ranking them as "More Severe." High initial cost of drones was ranked as the third most severe limitation, with 70% marking it as "More Severe.", Lack of literacy among farmers was 34 per cent (with 60% marking it as "Severe"), Volume of spray was seen as a limitation by 30% of respondents, with 50% marking it as "Severe."

Adverse weather conditions, such as rain, strong winds, and extreme temperatures, can significantly affect drone performance, reducing their efficiency and reliability (Smit and Patel, 2024) [4] and Zhang and Kovacs, 2012) [5]. The limited battery life and the time required for recharging remain a critical barrier to the widespread use of drones, particularly for large-scale agricultural operations. While drones offer long-term benefits, the upfront investment required for purchasing the drone and associated equipment can be a significant financial burden for many farmers, particularly those on smaller scales or in low-income areas. It's possible that lack education and technological know-how needed to completely comprehend and use drone technology. This restriction emphasizes how crucial it is to incorporate literacy programs, where hands-on instruction may enhance technical education and enable farmers to use drones efficiently. Although the use of drones to spray agrochemicals is growing, there are questions over their ability to manage high spray volumes, particularly for large size of land holdings. It could be difficult for farmers to apply enough fertilizer or pesticide across large regions in a single flight. This problem might be solved by enabling modular, attachable spray units or by increasing the drone's spray volume capabilities.

The study identifies a number of important obstacles that prevent drone technology from being widely used and used effectively in agriculture. The most urgent of these are the high initial cost, battery-related constraints, and weather-related problems. Manufacturers and legislators must address these issues right away in order to advance the technology and encourage its broad application. Other issues, such the requirement for further instruction, explanation, and assistance, imply that improving customer service infrastructure and educational initiatives are essential to improving drone use. In order to make drone technology more inclusive, more accessible and localized training as well as focused support programs are required, even though some of the limitations like the lack of literacy and locally relevant information are less severe.

Conclusion

Drone application in agricultural practices holds significant promise for transforming agriculture by offering innovative solutions for crop management, monitoring, and precision farming. Drones provide real-time aerial data that enhances decision-making, reduces costs, and improves efficiency. By enabling precise and timely interventions, drones contribute to better resource utilization, increased productivity, and sustainable farming practices. As drone technology continues to evolve, it is expected to become an integral part of modern agricultural practices, driving the future of precision agriculture and helping to meet the growing global food demand. Overall, fostering a supportive ecosystem that combines education, access, and financial incentives will empower farmers to influence drone technology effectively, ultimately enhancing productivity and sustainability in agriculture.

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